Python for computational Problem Solving · Unit_01 · Unit 1: Introduction

# DRAFT — UE26CS151A · Unit_01 · Unit 1: Introduction · Computer hardware

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

## Contents

1. 1 · Computer hardware
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

# Computer hardware

## 1 Computer hardware

Study focus: Computer hardware. The notes explain the key terms, distinctions, and practical method used in this lesson. Read the cited transcript intervals for source context, then use the examples to check your understanding. EV0000074 · EV0000075

SHORT-NOTE RECAP

**A digital computer processes encoded data through coordinated components.** The input unit supplies data and instructions; memory stores them; the processor fetches and executes instructions; and output devices present results. The arithmetic and logic unit performs calculations and comparisons, while the control unit coordinates instruction execution.

Registers and main memory are fast working storage but are volatile: their contents normally disappear when power is removed. Secondary storage such as an SSD retains files. The processor does not directly understand a human problem; software encodes the steps and data in forms the machine can process.

When tracing a program, follow both the changing values in memory and the sequence of instructions executed by the processor.

EV0000074 · EV0000075 · EV0000076 · +9 source units in pack.json

## A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Computer hardware transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/004_Computer_hardware__845ddeaeeec94ce6.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Computer hardware transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/004_Computer_hardware__845ddeaeeec94ce6.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Computer hardware transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/004_Computer_hardware__845ddeaeeec94ce6.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps. |
| Computer hardware transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/004_Computer_hardware__845ddeaeeec94ce6.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

### Known gaps, conflicts and corrections

1. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 16 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00022 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0054. Sidecar variants compare... |
| SRC00023 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0054. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00024 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0054. Sidecar variants compare... |
| SRC00025 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0054. Sidecar variants compare... |

# C Revision and retrieval practice